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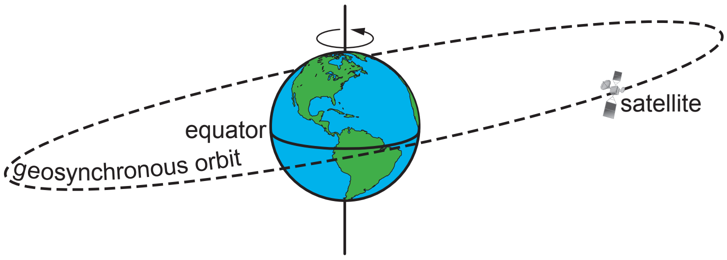
5. The diagram shows an example of a satellite in geosynchronous orbit around the Earth.

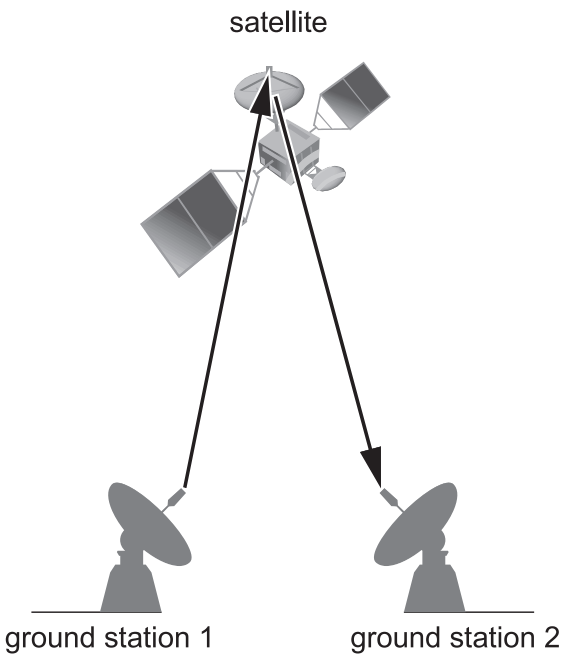


(a) State **one** way in which a geostationary orbit is different to the one shown above. [1]

....

....

(b) The diagram shows a signal sent from one ground station to a satellite which is then received back at a second ground station.



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(i) Use the equation:

$$\text{time} = \frac{\text{distance}}{\text{speed}}$$

to calculate the time interval between a microwave signal being sent from Earth to one of the satellites 36 000 000 m away and it being received back again. [3]
(speed of light, $c = 3 \times 10^8$ m/s)

time = ... s

(ii) Microwaves travel at the speed of light and have a range of wavelengths between 0.002 m and 1 m.
Use an equation from page 2 to calculate the **maximum** frequency of microwave signals. [3]

maximum frequency = ... Hz

(iii) Name **one** type of electromagnetic radiation which has frequencies smaller than those of microwaves. [1]

....

8

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